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Cryptology

2611006902

Theory Assignment - 6

Q1) Ciphertext ⇒ EVIRE

$P = (C - K) \bmod 26$

For all the 26 keys, possible solutions:

Keys	Plaintext	Keys	Plaintext
0	EVIRE	13	RIVER
1	DUHQD	14	GIHVDR
2	CIGPC	15	PGICP
3	BSFOB	16	OFSBO
4	ARENA	17	NERAN
5	ZQDMZ	18	MDQZM
6	YPCLY	19	LCPYL
7	XOBKX	20	KBOXK
8	VYNJW	21	JANWJ
9	VMZIV	22	IZNVI
10	ULVHU	23	HYLUH
11	TKXGT	24	GXKTG
12	SJWFS	25	FWJSF

when K=4,

E = (4-4) mod 26 = 0 → A

V = (21-4) mod 26 = 17 → R

I = (8-4) mod 26 = 4 → E

R = (17-4) mod 26 = 13 → N

E = (4-4) mod 26 = 0 → A

when K=13,

E = (4-13) mod 26 = 17 → R

V = (21-13) mod 26 = 8 → I

I = (8-13) mod 26 = 21 → V

R = (17-13) mod 26 = 4 → E

E = (4-13) mod 26 = 17 → R

Plaintext: ARENA

Plaintext: RIVER

a) what are the meaningful plaintexts Antony obtains?

Ans ARENA & RIVER

b) where should Antony meet Caesar?

Ans At the RIVER → TIBER RIVER
ARENA → COLOSSEUM

Both are meaningful; So Antony can't decode from the ciphertext alone.

c) what interesting point about a ciphertext-only attack do you learn from this exercise?

Ans A ciphertext only attack can give more than one meaningful plaintext when the ciphertext is short.

A longer ciphertext is needed for unique answer

Q2) Chosen - Plaintext Attack { use Affine Cipher (mod 26) }

HAHAHA → NONONO

H = 7 N = 13
A = 0 O = 14

Affine Cipher ⇒ $C = (k_1 \cdot P + k_2) \bmod 26$

$k_1 = a$; $k_2 = b$

b) $7a + b = 13 \mod 26$

$0a + b = 14 \mod 26$

$7a = -1 \mod 26$

$a = -(7^{-1}) \mod 26$

$a = -15 \mod 26$

$\boxed{a = 11}$

Then, $7(11) + b = 13 \mod 26$

$\Rightarrow b = -77 + 13 \mod 26$

$\Rightarrow b = -64 \mod 26$

$\Rightarrow \boxed{b = 14}$

c) $\boxed{a = 11}$ and $\boxed{b = 14}$

d) $C = 17(7) + 14 \mod 26$

$C = 13$

Q3) CR WW2; encrypted using Affine Cipher: mod 26. Attacker knows that plaintext begins with HA

a)

Plaintext	Ciphertext
H = 7	C = 2
A = 0	R = 17

Also; $C = (a \cdot P + b) \mod 26$

$7a + b = 2 \mod 26$ -①

$0a + b = 17 \mod 26$ -②

$7a = -15 \mod 26$

$a = 15 \quad (15 \bmod 26)$

$a = -225 \bmod 26$

$\Rightarrow \boxed{a = 9}$

then, $b = 17 \bmod 26$

$b = 17$

b) $C = (k_1 P + k_2) \bmod 26$

$C = a * P + 17 \bmod 26$

c) $P = k_1^{-1} \cdot (C - k_2) \bmod 26$

d) For W(22),

$P = k_1^{-1} (C - k_2) \bmod 26$

$= 9^{-1} (22 - 17) \bmod 26$

$= 3 \cdot (5) \bmod 26 \Rightarrow P = 15 \rightarrow P$

For Z(25),

$P = 9^{-1} (25 - 17) \bmod 26$

$= 3 \cdot 8 \bmod 26$

$\Rightarrow P = 24 \rightarrow Y$

Plaintext :— HAPPY

Q4) a) Plaintext : RIVER R = 17, I = 8, V = 21, E = 4, R = 17

NQXTP key:- N = 13, Q = 16, X = 23, T = 19, P = 15

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b) Encrypt

$C_i = (P_i + k_i - 1) \bmod 26$

R+N ⇒ $(17+13) \bmod 26 = 4 \rightarrow E$
I+Q ⇒ $(8+16) \bmod 26 = 24 \rightarrow Y$
V+X ⇒ $(21+23) \bmod 26 = 18 \rightarrow S$
E+T ⇒ $(4+19) \bmod 26 = 23 \rightarrow X$
R+P ⇒ $(17+15) \bmod 26 = 6 \rightarrow G$

Ciphertext : EYSXG

c) Decrypt

$P_i = (C_i - k_i) \bmod 26$

$(4-13, 24-16, 18-23, 23-19, 6-15) \bmod 26$
⇒ (17, 8, 21, 4, 17)
value corresponds to RIVER

d) Randomness:

A truly random key makes ciphertext unpredictable; patterns in a non-random key can reveal patterns in the message

Equal: Each plaintext character needs its own key character. A shorter key must be reused or repeated, which can create exploitable patterns

Using only one: Reusing a key lets attackers compare ciphertexts & potentially learn relationships between the plaintexts.